Grade 9 Science - Exploration

Chapter 1: Exploration - Entering the World of Secondary Science
Important Questions with Answers (38 Questions)

Section A: Multiple Choice Questions (1 mark each)

Q1. According to the chapter, science is not only about what we know, but also about:
(a) what we believe
(b) how we know it
(c) what we memorise
(d) what we inherit
Answer: (b) how we know it

Q2. In physics, a moving car in a simple model may be represented as:
(a) a sphere
(b) a layered structure
(c) a single point
(d) a diagram of parts
Answer: (c) a single point

Q3. In the cricket-shot model (Example 1.1), which detail is very important?
(a) Brand of the bat
(b) Colour of the ball
(c) Amount of grass on the field
(d) Speed and direction in which the ball is hit
Answer: (d) Speed and direction in which the ball is hit (along with the mass of the ball)

Q4. The symbol 'c' for the speed of light comes from the Latin word:
(a) celeritas
(b) cellula
(c) calor
(d) causa
Answer: (a) celeritas, meaning speed

Q5. Today, the speed of light is defined to be exactly:
(a) 3000 m/s
(b) 299792458 m/s
(c) 22300 m/s
(d) 2000 m/s
Answer: (b) 299792458 m/s

Q6. The passenger aircraft ran out of fuel mid-flight because of a mix-up between:
(a) litres and gallons
(b) metres and feet
(c) pounds (lb) and kilograms (kg)
(d) seconds and minutes
Answer: (c) pounds (lb) and kilograms (kg)

Q7. In science, a theory is:
(a) an untested guess
(b) an explanation based on careful testing and critical examination
(c) a personal opinion
(d) a final truth that never changes
Answer: (b) an explanation based on careful testing and critical examination

Q8. Applying the conservation of energy while climbing up the stairs is an example of a:

(a) law
(b) theory
(c) principle
(d) model

Answer: (c) principle

Q9. In the chapter, the compass on the page numbers reminds us that exploration needs:

(a) speed
(b) direction
(c) luck
(d) magnification

Answer: (b) direction - choosing appropriate models, asking the right questions and knowing the limits of our ideas

Q10. Roughly how many litres of air do we breathe in one day, as estimated in Example 1.3?

(a) 100 litres
(b) 1,000 litres
(c) 10,000 litres
(d) 10 lakh litres

Answer: (c) about 10,000 litres

Section B: True or False (1 mark each)

Q11. Ignoring certain details while building a model is a mistake made by scientists.

Answer: False. Ignoring details is done on purpose, to keep things simple enough while still answering the question.

Q12. Words like force, work, cell and reaction have specific meanings in science.

Answer: True. Science uses language carefully so that ideas are communicated clearly and unambiguously.

Q13. In science, a theory means a guess or an untested idea.

Answer: False. A theory is an explanation based on careful testing and critical examination.

Q14. Weather forecasts are usually reliable for a few hours or days, but less certain further into the future.

Answer: True. Tiny differences in conditions grow over time and lead to very different outcomes.

Q15. There is scientific evidence that food becomes harmful if eaten during an eclipse.

Answer: False. An eclipse is just a play of shadows; no physical, chemical or biological mechanism supports the claim.

Q16. The natural world is divided into strict, independent branches like physics, chemistry and biology.

Answer: False. These divisions are made by us to organise knowledge; nature has no such boundaries.

Section C: Fill in the Blanks (1 mark each)

Q17. The magnifying glass on the page numbers symbolises careful _____.

Answer: observation

Q18. Meghnad Saha treated the matter in a star as a hot _____.

Answer: gas

Q19. A _____ usually describes a regular pattern observed in nature.

Answer: law

Q20. At rest, we take about _____ breaths in a minute.

Answer: 12 - 15

Q21. Using _____ units everywhere avoids conversions and errors.

Answer: SI (standard)

Q22. When predictions do not match observations, scientists re-examine their assumptions, models or _____.

Answer: measurements

Section D: Assertion and Reason Questions (1 mark each)

Options: (A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion. (B) Both are true but Reason is not the correct explanation. (C) Assertion is true but Reason is false. (D) Assertion is false but Reason is true.

Q23. Assertion: Scientific models deliberately ignore certain details.
Reason: Ignoring details keeps things simple enough while still answering the question being asked.

Answer: (A) Both are true and the Reason explains the Assertion.

Q24. Assertion: A theory in science is not just a guess.
Reason: A theory is based on careful testing and critical examination of evidence.

Answer: (A) Both are true and the Reason explains the Assertion.

Q25. Assertion: Failure of a prediction is a strength of science.
Reason: When predictions fail, scientists re-examine assumptions, models or measurements instead of rejecting ideas on opinion or belief.

Answer: (A) Both are true and the Reason explains the Assertion.

Q26. Assertion: Weather forecasts become less certain for the distant future.
Reason: Very tiny differences in initial conditions can grow over time.

Answer: (A) Both are true and the Reason explains the Assertion.

Q27. Assertion: Mathematics in science is a language for thinking.
Reason: Mathematics is used only for finding numerical answers.

Answer: (C) Assertion is true but Reason is false. Mathematics helps us think clearly and reason carefully, not just calculate.

Section E: Short Answer Questions (2 marks each)

Q28. What is a model in science? Give one example from physics.

Answer: A model is a simplified way of looking at a real system that focuses only on what is most important for a given question. In physics, a moving car may be represented as a single point.

Q29. Why does science use a shared language of terms, symbols and units?

Answer: It allows scientists across the world to describe observations, compare results and build ideas together clearly and unambiguously. For example, mass, velocity, force and electric current are represented by m, v, F and I, each with a defined unit.

Q30. Differentiate between a law and a theory.

Answer: A law describes a regular pattern observed in nature, often in words or mathematical form (e.g. Newton's laws of motion). A theory goes a step further and explains why those patterns occur, based on evidence (e.g. the atomic theory explains how molecules are formed).

Q31. Why are scientific predictions not just guesses?

Answer: They are reasoned expectations based on evidence and careful thinking, using well-established laws, theories and models. When they match observations, confidence in the science grows; when they do not, scientists re-examine their ideas.

Q32. What caused the airplane fuel miscalculation incident, and what lesson does it teach?

Answer: The ground crew used the fuel density in pounds (lb) per litre instead of kilograms (kg) per litre, so the aircraft was about 15,000 litres short of fuel. The lesson is to use standard SI units everywhere to avoid conversion errors.

Q33. How can you scientifically test the claim that food becomes harmful during an eclipse?

Answer: Ask simple scientific questions: what physical change occurs in an eclipse? Does temperature change significantly? Does food go bad when left in a shadow? We conclude that an eclipse is only a play of shadows, so no physical, chemical or biological mechanism supports the claim.

Section F: Long Answer Questions (4 - 5 marks each)

Q34. Explain why scientists use models. Using the cricket-shot example, describe which details are kept and which are ignored.

Answer:

- The natural world is complex and cannot be studied in full detail, so science uses models - simplified ways of looking at real systems that focus on what matters for a given question.
- Examples: a car as a single point (physics); atoms as spheres and bonds (chemistry); cells as diagrams of key parts (biology); the Earth as a smooth layered sphere (earth science).
- Building a model involves making assumptions and deliberately ignoring details. These choices are not mistakes; they keep things simple but still let us answer our question.
- Cricket shot: the question is, 'Will the ball cross the boundary without hitting the ground first?' Important details are the mass of the ball and the speed and direction of the hit. Brand of the bat, colour of the ball and grass on the field make no difference. Air resistance, spin and seam stitching have small effects and can be ignored in a simple model.
- As we build more complex models, we add extra details for greater accuracy.

Q35. Describe the role of mathematics in science. Is it only a tool for calculation?

Answer:

- Mathematics is not meant to be a hurdle; it is a language that helps us think more clearly about the world.
- An equation is a compact statement about how certain things are related, not just a calculation tool.
- For example, distance, time and velocity let us predict where an object will be later. Mathematical expressions also describe rates of chemical reactions, population growth and energy changes.
- Using mathematics in science does not mean memorising equations. It means understanding the situation first, identifying the relevant quantities, and then using mathematical relationships to reason carefully.
- So mathematics is a powerful language for thinking, not just for finding numerical answers.

Q36. Estimate the volume of air you breathe in one day, showing all the steps (Example 1.3).

Answer:

- Breaths per day: at rest we take about 12 - 15 breaths a minute. There are 60 x 24 = 1440 minutes in a day, so about 18,000 - 22,000 breaths, roughly 20,000 breaths a day.
- Volume of one breath: about 4 - 5 breaths fill a rubber party balloon of about 2 litres, so one breath is about 0.5 litre.
- Total: 20,000 x 0.5 litre = about 10,000 litres of air a day.
- Cross-check: blowing a balloon takes about 20 s, so about 3 balloons a minute. (3 balloons/min) x (2 litres/balloon) x (1440 min/day) = 8640 litres, reasonably close to 10,000 litres.
- The aim of estimation is not an exact answer, but to check whether a result is reasonable. It builds intuition and helps detect errors.

Q37. 'Failures of predictions are the greatest strength of science.' Explain this statement with the help of weather forecasts.

Answer:

- Science can predict what will happen under new conditions using laws, theories and models. When predictions match observations, confidence in the science grows.
- When predictions do not match, scientists re-examine their assumptions, models or measurements. They reject ideas only on evidence, never on opinion or belief.
- Even the most successful theories have limits and may fail in new conditions or when measurements become more precise. No scientific theory is ever final or beyond question.
- This openness to being corrected by nature is what has helped science understand the world.
- Weather forecasts depend on temperature, pressure, humidity and wind. Very tiny differences in conditions grow over time, so forecasts are reliable for a few hours or days but less certain further ahead. Wrong forecasts lead to better measurements and models.

Q38. Explain how the study of science connects different branches of knowledge. Use the example of a mask.

Answer:

- After Grade 10, science is divided into branches such as physics, chemistry, biology and earth science, but the natural world has no such boundaries. These divisions are made by us to organise knowledge.
- Most real-world problems, such as climate change, developing medicines or designing sustainable technologies, need ideas from several disciplines together. Science also connects with mathematics, technology, arts and social sciences.
- Mask example: understanding how a mask works needs physics (particle motion and electrostatic attraction), chemistry (properties of polymer fibres), biology (size and behaviour of viruses) and mathematics (modelling airflow and filtration efficiency).
- Therefore, to make sense of the world fully, we need to connect multiple ways of knowing, each enriching the other.